Study Script

1. Welcome & Consent (~5 min)

👉 **Open and put in date and time**

Hi, thank you for taking the time to participate in our study!

My name is Lars, and this study is part of my master thesis. The title is *"Making Robot Thinking Visible"* and I am investigating how different ways of showing what a robot is thinking affects the user.

First i need you to fill out the consent form. Please take your time to read through it if you want.

👉 **Hand over consent form**

The study will be conducted in VR. I will give a short tutorial and overview how to use it. Then i also will have some interview sections which with your consent i will record the audio of so that i can transcribe it later.

The study takes about an hour in total. Your data will be anonymised and used only for research purposes. You can stop participating at any time, just let me know.

2. Pre-Study Questionnaire (~3 min)

Before we go into VR, I'd like you to fill out a short questionnaire on the laptop. It's just a few questions about your background.

👉 **Open Pre Study Questionnaire**

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3. VR Familiarisation (~3 min)

Now we'll get you set up in VR.

Take a moment to look around. You are in a kitchen, and in front of you is a PR2 robot. This is the robot you will be observing throughout the study as it prepares different recipes.

Let me walk you through how the interaction works:

1. First, the robot's thought bubble will show you what it is thinking about.
2. Then the simulation pauses and a menu appears with four options.
3. Based on what you saw in the thought bubble, you select what you think the robot is planning to do next
4. Then you can indicate your level of certainty on the scale.
5. After you make your choice, the robot performs its action.
6. After the action is completed, you need to assess whether the robot executed its plan successfully. This means determining whether the robot performed its plan as intended.

This will repeat for each step of the recipe.

3b. Baseline Block (~2.5 min)

Now you'll observe the robot preparing something short. This time there won't be a thought bubble — just watch the robot and make your predictions the same way. This will act as a baseline.

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4. Condition 1 (~8 min)

Great, now i will start the first main condition. This time the robot will prepare a recipe from A to Z and show a visualisation of its plan.

[Participant does Study]

Great, that was the first part. I'm going to ask you to take off the headset for a moment.

4b. Post-Condition 1 Elicitation (~2 min)

Before the questionnaire, I'd like to ask you three quick questions. I'll record your answers.

🎙️

1. **In your own words, describe how this robot decide what to do next?**
2. **Based on what you observed, what do you think this robot is good at, and where would it struggle?**
3. **If the robot encountered an unexpected obstacle, what do you think it would do?**

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5. Post-Condition 1 Questionnaire (~5 min)

Please fill out this questionnaire based on the robot you just observed.

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Group	Condition 1	Form	Condition 2	Form
1	Visual + Recipe A	**Form 1** (MDMT A)	Textual + Recipe B	**Form 2** (MDMT B)
2	Textual + Recipe B	**Form 1** (MDMT A)	Visual + Recipe A	**Form 2** (MDMT B)
3	Visual + Recipe B	**Form 2** (MDMT B)	Textual + Recipe A	**Form 1** (MDMT A)
4	Textual + Recipe A	**Form 2** (MDMT B)	Visual + Recipe B	**Form 1** (MDMT A)

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6. Short Break

The second part will work exactly the same way. This time however the robot will perform a different recipe and a different type of visualisation will appear.

Let's put the headset back on.

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7. Condition 2 (~8 min)

[Participant does Study]

7b. Post-Condition 2 Elicitation (~2 min)

Same three questions as before, but now about what you just experienced.

🎙️

1. **In your own words, describe how this robot decides what to do next.**
2. **Based on what you observed, what do you think this robot is good at, and where would it struggle?**
3. **If the robot encountered an unexpected obstacle, what do you think it would do?**

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8. Post-Condition 2 Questionnaire (~5 min)

Same questionnaire as before, but this time based on the second experience you just had.

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Group	Condition 1	Form	Condition 2	Form
1	Visual + Recipe A	**Form 1** (MDMT A)	Textual + Recipe B	**Form 2** (MDMT B)
2	Textual + Recipe B	**Form 1** (MDMT A)	Visual + Recipe A	**Form 2** (MDMT B)
3	Visual + Recipe B	**Form 2** (MDMT B)	Textual + Recipe A	**Form 1** (MDMT A)
4	Textual + Recipe A	**Form 2** (MDMT B)	Visual + Recipe B	**Form 1** (MDMT A)

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9. Post-Study Interview (~10 min)

Now finally I'd like to ask you a few questions about your experience.

Post-Study Interview

Conducted without headset. Audio recorded.

1. **A: What part of the first condition stuck with you? What did you see in the thought bubble and what went through your mind?**
   B: And for the second condition?
3. **Did you develop a strategy for making predictions and did it differ between conditions?**
4. **In which condition did you feel like you understood the robot better and in which do you think you actually predicted its actions more accurately?**
5. **Which was more mentally demanding to interpret and which one would you prefer to interact with?**
6. **Were there moments where you realized the robot works differently than you expected? What changed in your understanding?**
7. **If you were actually working alongside this robot, would the thought bubble have changed how you coordinated with it?**
8. **Were you familiar with any of these recipes beforehand?** cranberry fluff salad and mexican wedding cookies
9. **Anything you would like to add?**

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10. Wrap-Up (~2 min)

That's it — we're done! Thank you very much for participating.